

Postulates of Quantum Mechanics 1

The state of a system is completely specified by its wave-function $\psi(x)$ that depends on coordinate of the particle. All possible information can be derived from $\psi(x)$. This function is called wave function, has the important property that $\psi^*(x) \psi(x) dx$ is the probability that the particle lies in the interval dx , located at the position x .

$$\psi^*(\vec{r}, t) \psi(\vec{r}, t) dx dy dz = \left(\begin{array}{l} \text{Probability particle is in} \\ \text{volume element } dx dy dz \\ \text{at } \vec{r} \text{ at } t \end{array} \right)$$

(Measurable properties) $\rightarrow$ observables $\{$ Classical

If we know $\psi(\vec{r}, t) \rightarrow$ we have all observable in Q.M

If we have $\vec{r}, \vec{p}$ $\rightarrow$ we know $\vec{r}(t)$ | C.M

and Force

trajectories

$$\iint_{\text{all space}} \psi^*(\vec{r}, t) \psi(\vec{r}, t) = 1$$

$$\int_{\mathbb{R}^3} d\tau \psi^*(\vec{r}, t) \psi(\vec{r}, t) = 1$$

$$d\tau = dx dy dz$$

$$\int_{\mathbb{R}^n} \rightarrow \int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} n$$

$\psi(\vec{r}, t)$ must be

"normalized"

"Square

Integrable"

$$\int_{\mathbb{R}} [f(x)]^2 dx = \text{Constant}$$

$$|C| < \infty$$

Postulate 2 of Quantum Mechanics

For every classical observables, there corresponds a linear, quantum mechanical operator.

Linear: $\hat{O} \left(\sum_{i=1}^n c_i \psi_i \right) = \sum_{i=1}^n c_i \hat{O} \psi_i \quad \{c_i\} \in \mathbb{C}$

x^n ✓ $\frac{d^n}{dx^n}$ ✓ $\int () dx$ ✓ $\sqrt{x()^2 x}$

Position operator — will explain later.

Observable	Symbol	Operator
Total Energy	E	$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V(x, y, z)$
Potential Energy	V	$\hat{V} = V(x, y, z)$
Kinetic Energy	T	$\hat{T} = \frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$
Position	x $\vec{r}$	$\hat{x} = x$ $\vec{\hat{r}} = \hat{x}\hat{i} + \hat{y}\hat{j} + \hat{z}\hat{k}$
Momentum	p_x $\vec{p}$	$\hat{p}_x = -i\hbar \frac{d}{dx}$ $\vec{\hat{p}} = -i\hbar \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)$

For more check

Page 119, McQuarrie

Postulate 3

In any measurement of observable associated with the operator $\hat{A}$, the only values that will ever be observed are eigenvalues a_n , which satisfy the eigenvalue equation

$$\hat{A}\psi_n = a_n\psi_n$$

Ex. Schrödinger equation: $\hat{H}\psi_n = E_n\psi_n$

↑ "Eigen Function"

↑ "Eigen value"

ψ_n
 E_n

Postulate 4

If a system is in a state described by normalized wave function ψ , then the average value of the observable corresponding to $\hat{A}$ is given by

$$\langle a \rangle = \int_{\text{all space}} \psi^* \hat{A} \psi dx$$

Bra-ket notation
or Dirac notation

Suppose that $\psi(x)$ is eigenfunction of $\hat{A}$ where $\hat{A} \psi_n(x) = a_n \psi_n(x)$

$$\begin{aligned} \text{Then } \langle a \rangle &= \int_{-\infty}^{\infty} \psi_n^*(x) \hat{A} \psi_n(x) dx = \int_{-\infty}^{\infty} \psi_n^*(x) a_n \psi_n(x) dx \\ &= a_n \int_{-\infty}^{\infty} \psi_n^*(x) \psi_n(x) dx = a_n \end{aligned}$$

Furthermore, if $\hat{A} \psi_n(x) = a_n \psi_n(x)$, then

$$\hat{A}^2 \psi_n(x) = \hat{A} [\hat{A} \psi_n(x)] = \hat{A} [a_n \psi_n(x)] = a_n \hat{A} \psi_n(x)$$

$\downarrow$
 $a_n [\hat{A} \psi_n(x)]$
 $\downarrow$
 $a_n \psi_n(x)$

Variance of two measurements is -

$$\sigma_a^2 = \langle a^2 \rangle - \langle a \rangle^2 = a_n^2 - a_n^2 = 0$$

Thus, as postulate 3 says, the only value we measure is $\rightarrow a_n$

Postulate 5 of Quantum mechanics

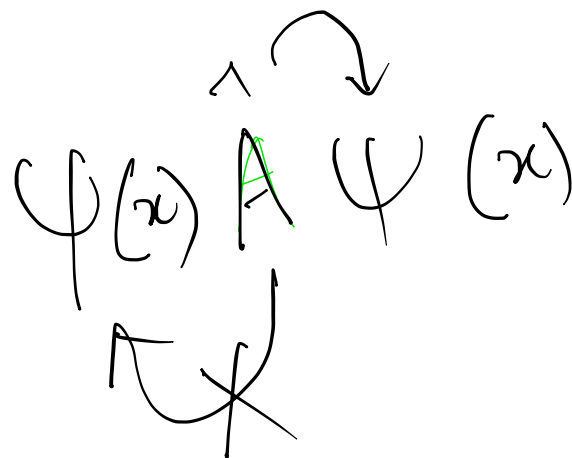
The wave function, or state of function, of a system evolves in time according to the time-dependent Schrödinger eqn.

$$\hat{H}\Psi(x,t) = i\hbar \frac{\partial \Psi(x,t)}{\partial t}$$

Solution : $\Psi_n(x,t) = \psi_n(x) e^{-iE_n t/\hbar}$

ψ

From time-independent

$$\psi(x) \hat{A} \psi(x)$$


A handwritten diagram showing the expression $\psi(x) \hat{A} \psi(x)$. A curved arrow points from the operator $\hat{A}$ to the second $\psi(x)$ term. A second curved arrow, which is crossed out with a large 'X', points from the first $\psi(x)$ term to the operator $\hat{A}$.

$$\hat{A}^2 \psi(x)$$